Digital Communication IC Design

Homework 3

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Development Environment

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Setup and Code Organization

My student ID is M11407439, so $m = 39$. This number is only used as the random seed in Q2 and Q4 (`rng(39, 'twister')`). Everything else in Q1 and Q3 follows from the power delay profile and $F_s = 250$ MHz.

Table 1.1: Files in `01_src`

File	Content
`hw3_channel.m`	PDP, p_i, $\tilde{p}_i$, n_i, and the seed-39 gains γ_i, g_i, h_i, $h[n]$
`ofdm_size.m`	Chooses N_g and N from the channel length (Q3)
`q1.m` – `q4.m`	One script per question; prints the numbers and saves the figures
`new_fig.m`, `save_fig.m`, `tl_labels.m`, `hw_palette.m`	Figure style only

All four scripts call `hw3_channel.m`, so Q2, Q3 and Q4 use exactly the same channel realization. The numbers in this report are copied from the console output of the scripts.

Q1 — Power Delay Profile Statistics

Consider the four-path channel power delay profile below. Use linear power, not dB values, when calculating channel statistics.

Tap i	Power $P_{i,\mathrm{dB}}$	Delay τ_i
0	0 dB	0 ns
1	−2 dB	16 ns
2	−6 dB	92 ns
3	−9 dB	200 ns

2.1 (a) Linear and Normalized Powers

Convert every path power to linear scale, normalize the four powers so that they sum to one, and print both vectors, $\mathbf{p} = [p_1\ p_2\ p_3\ p_4]$ and $\tilde{\mathbf{p}} = [\tilde{p}_1\ \tilde{p}_2\ \tilde{p}_3\ \tilde{p}_4]$ with at least six significant digits. (10%)

$$p_i = 10^{P_{i,\mathrm{dB}}/10}, \qquad \tilde{p}_i = \frac{p_i}{\sum_r p_r}$$

The table numbers the paths from 0 to 3 while the vectors in the question are written from 1 to 4. They describe the same four paths, so I keep the tap index $i = 0, \ldots, 3$ everywhere; vector entry j is tap $j - 1$. `q1.m` prints

$$\mathbf{p} = [\,1.00000000 \quad 0.63095734 \quad 0.25118864 \quad 0.12589254\,], \qquad \textstyle\sum_r p_r = 2.0080385,$$
$$\tilde{\mathbf{p}} = [\,0.49799841 \quad 0.31421576 \quad 0.12509155 \quad 0.06269429\,], \qquad \textstyle\sum_r \tilde{p}_r = 1.$$

The first path is the 0 dB reference, so $p_0 = 1$ exactly. After normalization it carries almost half of the power, and the first two paths together hold 81% of it.

2.2 (b) Mean Excess Delay, RMS Delay Spread and Maximum Excess Delay

Calculate the mean excess delay, RMS delay spread, and maximum excess delay. Show the contribution of each path to the first and second delay moments. (10%)

$$\bar{\tau} = \sum_i \tilde{p}_i \tau_i, \qquad \tau_{\mathrm{RMS}} = \sqrt{\sum_i \tilde{p}_i (\tau_i - \bar{\tau})^2}$$

Table 2.1 lists what each path adds to the first moment $\sum_i \tilde{p}_i \tau_i$, to the second raw moment $\sum_i \tilde{p}_i \tau_i^2$, and to the second central moment $\sum_i \tilde{p}_i (\tau_i - \bar{\tau})^2$.

Table 2.1: Power Delay Profile and the Contribution of Each Path to the Delay Moments

i	$P_{i,\mathrm{dB}}$	p_i	$\tilde{p}_i$	τ_i (ns)	$\tilde{p}_i \tau_i$ (ns)	$\tilde{p}_i \tau_i^2$ (ns^2)	$\tilde{p}_i (\tau_i - \bar{\tau})^2$ (ns^2)
0	0 dB	1.00000	0.497998	0	0.0000	0.00	420.98
1	−2 dB	0.630957	0.314216	16	5.0275	80.44	53.71
2	−6 dB	0.251189	0.125092	92	11.5084	1058.77	495.31
3	−9 dB	0.125893	0.0626943	200	12.5389	2507.77	1831.64
Sum		2.00804	1.00000		29.0747	3646.99	2801.65

From the sums in the last row,

$$\bar{\tau} = \sum_i \tilde{p}_i \tau_i = 29.0747 \text{ ns}, \tag{2.1}$$

$$\tau_{\mathrm{RMS}} = \sqrt{\sum_i \tilde{p}_i (\tau_i - \bar{\tau})^2} = \sqrt{2801.65 \text{ ns}^2} = 52.9306 \text{ ns}, \tag{2.2}$$

$$\tau_{\max} = \tau_3 - \tau_0 = 200 \text{ ns}. \tag{2.3}$$

The raw moments give the same spread, $\sqrt{\overline{\tau^2} - \bar{\tau}^2} = \sqrt{3646.99 - 29.0747^2} = 52.9306$ ns. For the maximum excess delay I take the last path relative to the first arriving one. All four paths are within 9 dB of the strongest, so the common −10 dB threshold would not drop any of them and the answer stays 200 ns.

The bar charts in the figure below make one thing clear. The strongest path sits at $\tau = 0$ and adds nothing to $\bar{\tau}$, while the weakest one, at 200 ns with $\tilde{p}_3 = 0.063$, adds the most (12.54 ns). In the second central moment the distance to $\bar{\tau}$ is squared, so the late path matters even more: it gives 1832 of the 2802 ns^2, about 65%, with only 6% of the power. That weak echo is the main reason τ_{RMS} comes out larger than $\bar{\tau}$.

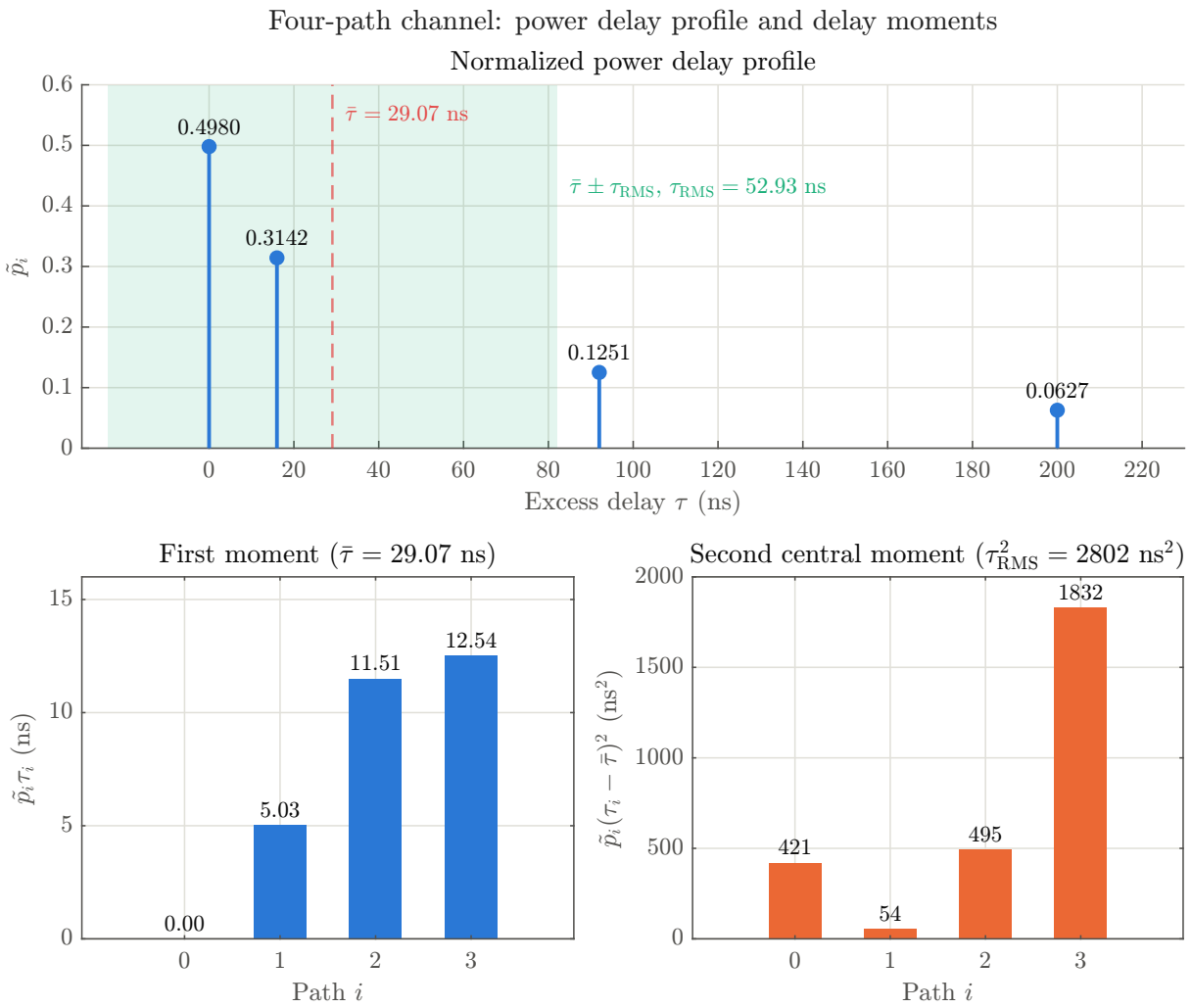


Figure 2.1: Normalized PDP with $\bar{\tau}$ (red) and $\bar{\tau} \pm \tau_{\mathrm{RMS}}$ (green), and the Contribution of Each Path to the First and Second Central Moments

2.3 (c) Coherence Bandwidth

Using the definition below, calculate the coherence bandwidth. Compare it with signal bandwidths of 2.5 MHz and 250 MHz and classify each channel as approximately flat fading or frequency selective. Explain the criterion. (5%)

$$B_c = \frac{1}{5\tau_{\mathrm{RMS}}}$$

$$B_c = \frac{1}{5 \times 52.9306\ \mathrm{ns}} = 3.7785\ \mathrm{MHz}. \tag{2.4}$$

Table 2.2: Signal Bandwidth Compared with $B_c = 3.78$ MHz

B_s	B_s/B_c	Symbol time $1/B_s$	Classification
2.5 MHz	0.66	400 ns $\approx 7.6\ \tau_{\mathrm{RMS}}$	approximately flat fading
250 MHz	66.2	4 ns $\approx 0.076\ \tau_{\mathrm{RMS}}$	frequency selective

The criterion I use is the one in the course slides (Ch. 4, p. 25). B_c is roughly the frequency separation over which the channel response stays correlated. When the signal bandwidth is below B_c, all of the signal sees about the same complex gain, and the channel just scales and rotates it: flat fading. When $B_s > B_c$, different parts of the band get different gains, and part of the band can land in a deep null: frequency-selective fading. Seen in time, a symbol much longer than the delay spread is flat fading, and a symbol that is not much longer than the delay spread suffers inter-symbol interference.

For 2.5 MHz, $B_s < B_c$, so I classify the channel as flat fading. I would call it "approximately" flat, as the question does. The ratio is 0.66, not $\ll 1$, and $K = 5$ is the most lenient value in the range $K = 5 \sim 50$ from the slides. With $K = 50$ we would get $B_c = 378$ kHz, and the same signal would count as frequency selective. So this case sits near the boundary, and the gain will still tilt slightly across the 2.5 MHz band.

For 250 MHz, B_s is 66 times B_c and the channel is clearly frequency selective. The band holds many fades (the nulls of $|H(f)|$ in Q2(b) are only a few MHz apart), and a 4 ns single-carrier symbol would be smeared over about 50 symbols by the 200 ns channel. Q3 and Q4 handle this case with OFDM.

2.4 (d) Integer Sample Delays at $F_s = 250$ MHz

At sampling rate $F_s = 250$ MHz, convert every path delay to an integer sample delay. Determine the discrete channel length and the minimum cyclic-prefix length that prevents inter-symbol interference for this channel. (5%)

The sample period is $T_s = 1/F_s = 4$ ns, and every delay is a multiple of it:

$$n_i = \tau_i F_s = [\,0,\ 16,\ 92,\ 200\,]\ \mathrm{ns} \times 250\ \mathrm{MHz} = [\,0,\ 4,\ 23,\ 50\,]. \tag{2.5}$$

No rounding is needed; the largest gap between $\tau_i F_s$ and the integer is 7×10^{-15}, which is just floating-point error.

The discrete channel $h[n]$ has nonzero taps at $n = 0, 4, 23, 50$, so its length is

$$L = \max_i n_i + 1 = 51\ \mathrm{taps}. \tag{2.6}$$

Convolution with $h[n]$ spreads each transmitted sample over the following $L - 1 = 50$ samples. The cyclic prefix has to absorb this spread, so the minimum CP length is

$$N_{\mathrm{CP,min}} = L - 1 = 50 \text{ samples} = 200 \text{ ns} = \tau_{\max}. \tag{2.7}$$

Q4(c) confirms that 50 is the exact limit: with a 49-sample CP the channel estimate is already wrong.

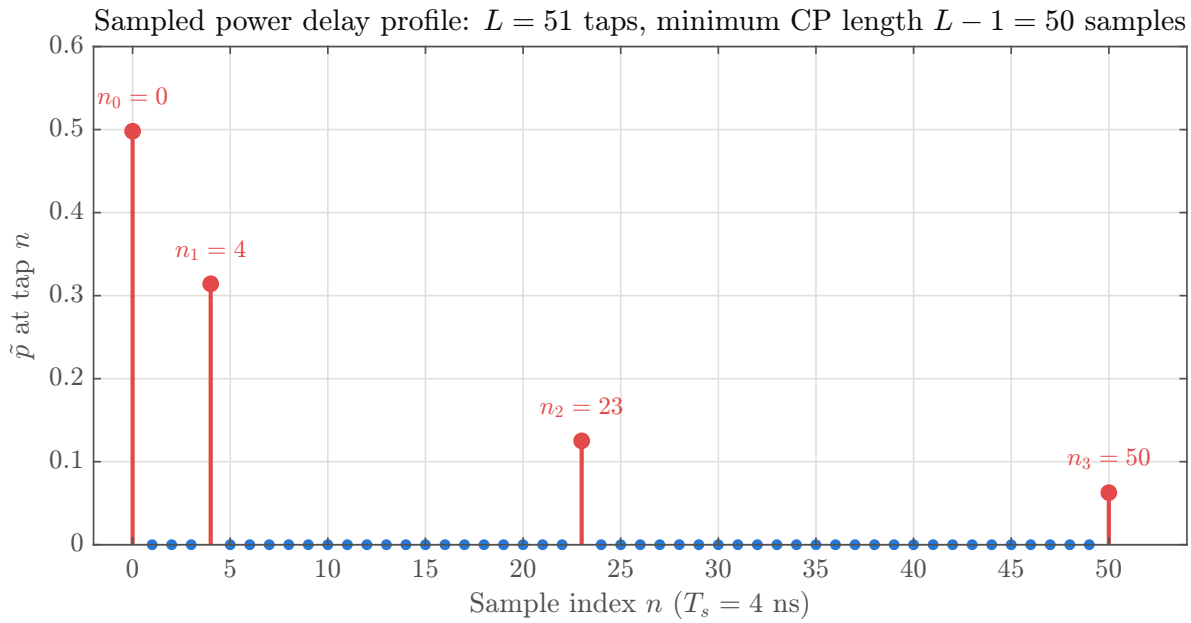


Figure 2.2: Power Delay Profile on the $F_s = 250$ MHz Sample Grid

Q2 — Random Channel and Its Frequency Response

Let m be the last two digits of the student ID and use m as the random seed. Generate independent γ_i distributed as $\mathcal{CN}(0,1)$ (complex normal distribution), with real and imaginary components each having variance 1/2. Use the powers and delays from Question 1.

3.1 (a) Complex Path Gains

Generate the four complex path gains and normalize them using the equation below. Print γ_i, g_i, h_i, their magnitudes and phases, and the final sum of squared magnitudes. (15%)

$$g_i = \sqrt{p_i}\,\gamma_i, \qquad h_i = \frac{g_i}{\sqrt{\sum_r |g_r|^2}}$$

With `rng(39, 'twister')` I draw `gamma = (randn(1,4) + 1j*randn(1,4))/sqrt(2)`, so the real and imaginary parts are independent $\mathcal{N}(0,1/2)$ and $E|\gamma_i|^2 = 1$. Table 3.1 lists the results (the console prints six decimals; four are kept here).

Table 3.1: Path Gains γ_i, g_i and h_i with Their Magnitudes and Phases

i	γ_i		$g_i = \sqrt{p_i}\,\gamma_i$		h_i	
	value	$\lvert\cdot\rvert, \angle\cdot$ (rad)	value	$\lvert\cdot\rvert, \angle\cdot$ (rad)	value	$\lvert\cdot\rvert, \angle\cdot$ (rad)
0	$0.0190 + 0.2488j$	$0.2495, 1.4947$	$0.0190 + 0.2488j$	$0.2495, 1.4947$	$0.0203 + 0.2664j$	$0.2672, 1.4947$
1	$0.9222 + 0.0498j$	$0.9235, 0.0540$	$0.7325 + 0.0396j$	$0.7336, 0.0540$	$0.7846 + 0.0424j$	$0.7857, 0.0540$
2	$0.5923 - 0.0540j$	$0.5947, -0.0909$	$0.2968 - 0.0271j$	$0.2981, -0.0909$	$0.3180 - 0.0290j$	$0.3193, -0.0909$
3	$-1.2012 - 0.0768j$	$1.2037, -3.0777$	$-0.4262 - 0.0273j$	$0.4271, -3.0777$	$-0.4565 - 0.0292j$	$0.4575, -3.0777$
$\sum \lvert\cdot\rvert^2$	2.717686		0.871620		1.000000	

The final sum is $\sum_i |h_i|^2 = 1.000000000000000$.

Multiplying by $\sqrt{p_i} > 0$ and dividing by a positive constant only changes magnitudes, so γ_i, g_i and h_i share the same phase. The question uses p_i rather than $\tilde{p}_i$ inside g_i. This does not change h_i: $\tilde{p}_i = p_i / \sum_r p_r$ only adds a common factor, and the normalization removes it again.

The powers of this realization, $|h_i|^2 = [0.071,\ 0.617,\ 0.102,\ 0.209]$, are quite different from the average profile $\tilde{p}_i = [0.498,\ 0.314,\ 0.125,\ 0.063]$. Path 0 happened to fall into a fade ($|\gamma_0| = 0.25$) and path 3 got a large draw ($|\gamma_3| = 1.20$). This is expected for Rayleigh fading. The PDP only fixes $E|h_i|^2$, and a single snapshot can look quite different from the average.

3.2 (b) Continuous Frequency Response

The continuous-time impulse response is given as follows. Derive its continuous-frequency response. Draw the frequency domain response including magnitude and phase versus frequency. Note that the frequency range should cover −250 MHz to 250 MHz. (15%)

$$h(t) = \sum_{i=0}^{3} h_i \, \delta(t - \tau_i)$$

A delayed impulse transforms into a linear-phase exponential, $\mathcal{F}\{\delta(t - \tau_i)\} = e^{-j2\pi f \tau_i}$, and the Fourier transform is linear, so

$$H(f) = \int_{-\infty}^{\infty} h(t)\, e^{-j2\pi f t}\, dt = \sum_{i=0}^{3} h_i \, e^{-j2\pi f \tau_i} = h_0 + h_1 e^{-j2\pi f (16\,\text{ns})} + h_2 e^{-j2\pi f (92\,\text{ns})} + h_3 e^{-j2\pi f (200\,\text{ns})}. \tag{3.1}$$

Writing each gain in polar form, $h_i = |h_i| e^{j\theta_i}$ with $\theta_i = \angle h_i$ from Table 3.1, the real and imaginary parts are

$$\operatorname{Re} H(f) = \sum_{i=0}^{3} |h_i| \cos(\theta_i - 2\pi f \tau_i), \qquad \operatorname{Im} H(f) = \sum_{i=0}^{3} |h_i| \sin(\theta_i - 2\pi f \tau_i), \tag{3.2}$$

so the magnitude and phase plotted below are

$$|H(f)| = \sqrt{\left(\operatorname{Re} H(f)\right)^2 + \left(\operatorname{Im} H(f)\right)^2}, \qquad \angle H(f) = \operatorname{atan2}\left(\operatorname{Im} H(f),\, \operatorname{Re} H(f)\right) \in (-\pi, \pi]. \tag{3.3}$$

I evaluate them on a 50 kHz grid from −250 to 250 MHz.

$H(f)$ is a sum of four phasors that turn at different rates as f changes. Where they line up, $|H(f)|$ reaches 1.83; where they cancel, it drops to 0.007 (−43 dB, at −43.95 MHz). The fastest phasor belongs to the 200 ns path and makes one turn every $1/200$ ns = 5 MHz, which sets the fine ripple in the plot. Near each deep null the phase jumps by almost π, because $H(f)$ passes close to the origin. Between nulls the phase falls steadily with f, as the $e^{-j2\pi f \tau_i}$ terms predict.

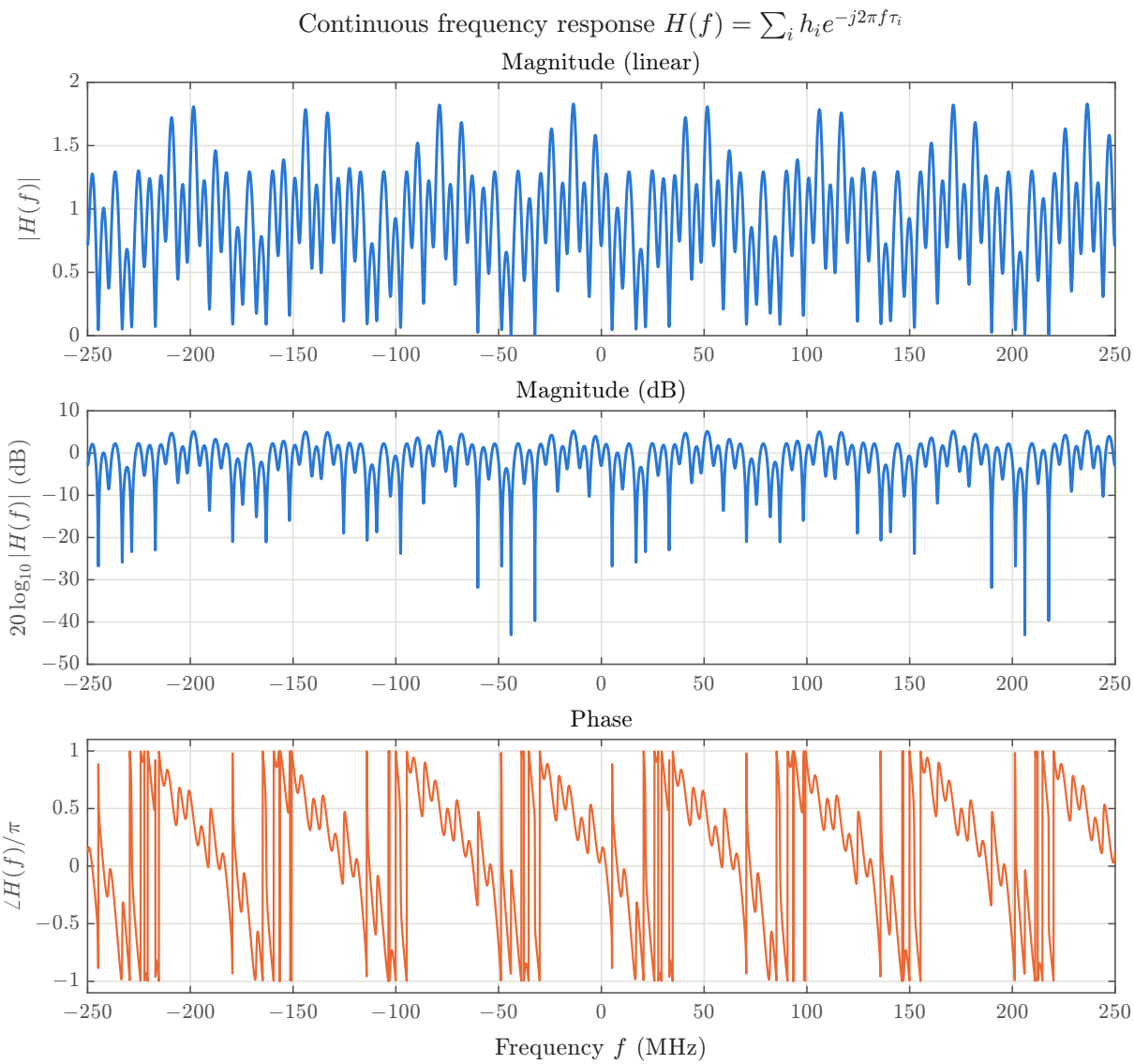


Figure 3.1: Magnitude (Linear and dB) and Phase of $H(f)$ from -250 to 250 MHz

3.3 (c) Verification and Periodicity

How do you verify your figure in Q2(b)? Do you observe any periodicity? Please state the reasons for the periodicity or non-periodicity of the figure. (10%)

Verification. I checked the figure with an independent computation and with two values I can predict by hand.

Every τ_i is a whole number of samples ($\tau_i = n_i T_s$), so $H(f) = \sum_i h_i e^{-j2\pi f n_i / F_s}$ is exactly the DTFT of $h[n]$ at $\omega = 2\pi f / F_s$. A 200-point FFT of $h[n]$ samples that DTFT every 1.25 MHz, without using the closed-form expression at all. In the figure below the FFT points (circles) sit on the closed-form curve in both magnitude and phase, and the largest difference over 401 points is 1.5×10^{-14}.

At $f = 0$ every exponential equals 1, so $H(0) = \sum_i h_i = 0.6664 + 0.2506j$; the plot gives $|H(0)| = 0.712$, as it should. Parseval provides the second check. Averaged over one period, the cross terms $h_i h_r^* e^{-j2\pi f(\tau_i - \tau_r)}$ vanish and $\overline{|H(f)|^2} = \sum_i |h_i|^2 = 1$. The numerical mean is 1.000000, which is also why $|H(f)|$ hovers around 1 in the plot. The bottom panel is a third check, on periodicity, discussed next.

Periodicity. Yes. The curve from -250 to 0 MHz repeats exactly from 0 to 250 MHz, in magnitude and in phase, so $H(f)$ is periodic with period 250 MHz. The bottom panel overlays $|H(f)|$ on $[-250, 0]$ with $|H(f + 250\ \text{MHz})|$; the two curves coincide, with a largest difference of 2.0×10^{-14}.

The reason is that all delays are integer multiples of a common delay. $H(f + P) = H(f)$ for every f requires $e^{-j2\pi P\tau_i} = 1$ for every path, i.e. $P\tau_i$ must be an integer for all i. The largest delay unit shared by all paths is

$$\gcd(16, 92, 200)\ \text{ns} = 4\ \text{ns} = T_s \qquad \Rightarrow \qquad P = \frac{1}{\gcd(\tau_i)} = \frac{1}{4\ \text{ns}} = 250\ \text{MHz} = F_s. \tag{3.4}$$

250 MHz is also the *smallest* period. In samples the delays are 4, 23 and 50, and $\gcd(4, 23, 50) = 1$, so no shorter P makes all three products integers. The script checks this as well: shifting by 125 MHz or 62.5 MHz leaves errors of 0.64 and 1.37. The same holds for $|H(f)|$ alone, since $|H(f)|^2$ contains the differences $\tau_i - \tau_r$, whose gcd is again 4 ns.

From the sampling point of view this is the familiar rule that a discrete-time signal has a 2π-periodic spectrum. All taps lie on the 4 ns grid, so the channel acts like the discrete filter $h[n]$ and its response repeats every F_s. The required range of ± 250 MHz therefore shows exactly two periods. If one delay were off the common grid (say an irrational ratio such as 16 ns and $10\sqrt{2}$ ns), no common period would exist and $H(f)$ would never repeat exactly.

What the figure does *not* have is mirror symmetry: $|H(-f)| \neq |H(f)|$, with differences up to 0.62. A real impulse response would give $H(-f) = H^*(f)$, but the baseband gains h_i are complex, so that symmetry does not apply.

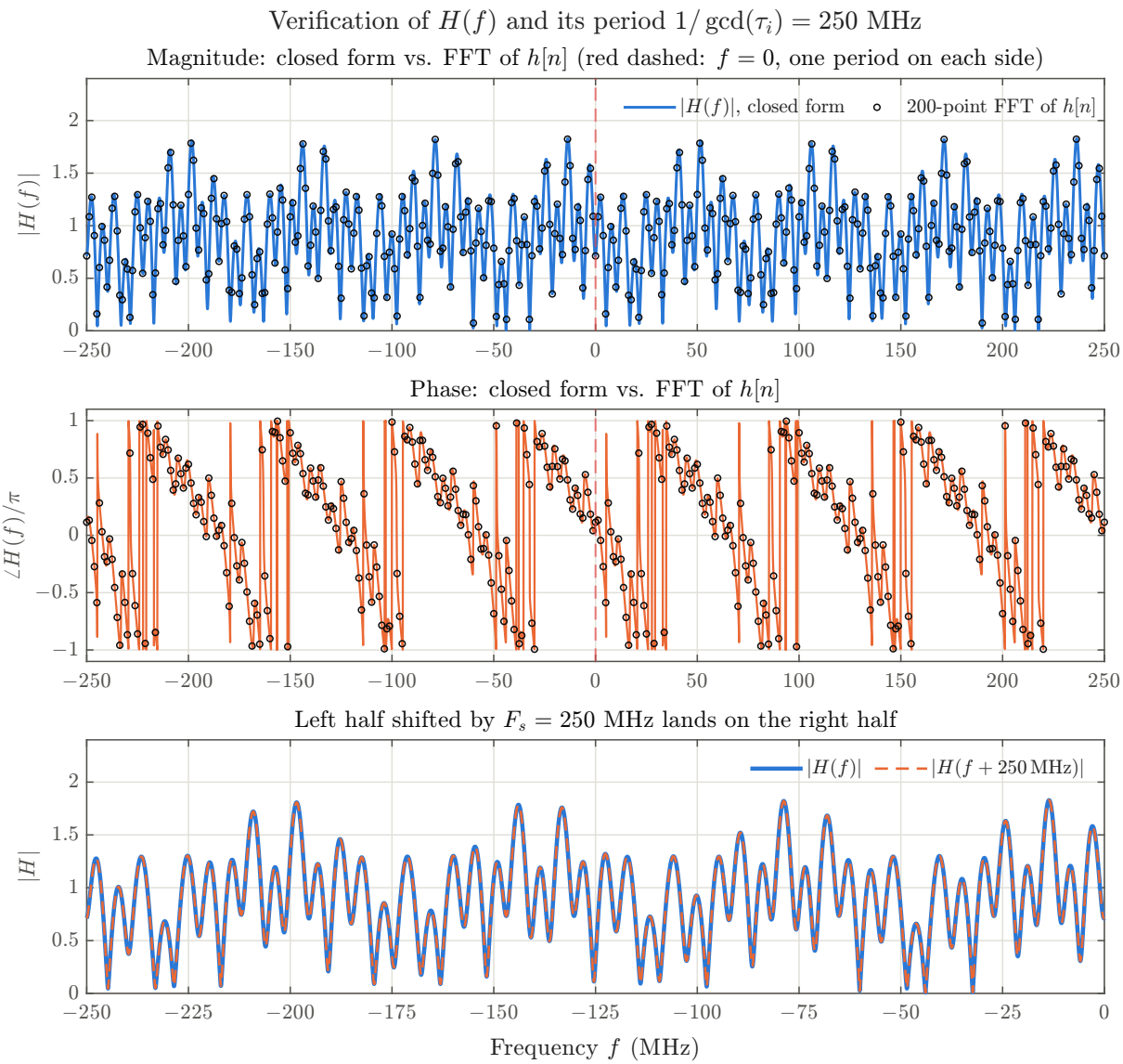


Figure 3.2: Verification of $H(f)$: Closed Form vs. FFT of $h[n]$ (top, middle) and $|H(f)|$ vs. $|H(f+250\,\text{MHz})|$ (bottom)

Q3 — OFDM System Design

Use F_s = 250 MHz and the time-invariant discrete channel from Question 2 to design an OFDM system.

4.1 (a) Guard Interval

Determine the minimum guard interval length to prevent ISI, and select an appropriate N_g value that is a power of 2. (10%)

The discrete channel from Q2 is $h[n] = \sum_i h_i\,\delta[n - n_i]$ with $n_i = 0, 4, 23, 50$, so its memory is $L - 1 = 50$ samples. The guard interval must cover this memory; otherwise the tail of one OFDM symbol runs into the useful part of the next one:

$$N_{g,\min} = L - 1 = 50 \text{ samples} = 200 \text{ ns}. \tag{4.1}$$

The powers of 2 next to 50 are 32 and 64. 32 is too short, so I choose

$$N_g = 64 \text{ samples} = 256 \text{ ns}, \tag{4.2}$$

which leaves 14 samples (56 ns) of margin, for example against timing errors.

4.2 (b) FFT Size

If the guard interval ratio should be smaller than or equal to 1/4, what is the proper setting of FFT size N in an OFDM system. Select the minimum value to satisfy the constraint. (5%)

(The original wording said "smaller than 1/4"; the announced correction changed it to "smaller than or equal to 1/4", which is what I follow here.) $N_g/N \le 1/4$ means $N \ge 4N_g = 256$, and the FFT size is a power of 2:

Table 4.1: Guard Interval Ratio for $N_g = 64$

N	N_g/N	$\le 1/4$?
128	1/2	no
256	**1/4**	**yes (equal is allowed)**
512	1/8	yes, but not the minimum

$N = 128$ gives 1/2, which is too large. $N = 256$ gives exactly 1/4, and since the constraint now allows equality, 256 is the minimum FFT size:

$$N = 256, \qquad \frac{N_g}{N} = \frac{1}{4}, \qquad \Delta f = \frac{F_s}{N} = 976.5625 \text{ kHz}. \tag{4.3}$$

One OFDM symbol is $N + N_g = 320$ samples: 1024 ns of useful time plus a 256 ns CP, so the CP costs $64/320 = 20\%$ of the air time. The subcarrier spacing of 977 kHz is about 3.9 times smaller than $B_c = 3.78$ MHz from

Q1(c), so each subcarrier on its own still sees a nearly flat channel, even though the full 250 MHz band is strongly frequency selective.

4.3 (c) Channel Response at the Subcarriers

Now calculate the subcarrier frequency f_k in the frequency domain for $k = -N/2 \sim N/2 - 1$. Draw the frequency domain response at each subcarrier of the channel described in Q2. Note that the frequency response should include both phase and magnitude. Please use stem plots. (10%)

The subcarrier frequencies are

$$f_k = k\,\Delta f = k\,\frac{F_s}{N}, \qquad k = -128, \ldots, 127, \qquad f_k \in [-125,\ 124.02]\ \text{MHz}, \tag{4.4}$$

and the channel response at each of them follows from Q2(b):

$$H[k] = H(f_k) = \sum_{i=0}^{3} h_i\, e^{-j2\pi f_k \tau_i} = \sum_{i=0}^{3} h_i\, e^{-j2\pi k n_i/N}. \tag{4.5}$$

The second form shows that $H[k]$ is also the N-point DFT of $h[n]$, with k reordered to $-N/2 \sim N/2 - 1$. The script finds the two equal to within 7.9×10^{-15}. Table 4.2 lists some of the 256 subcarriers: both band edges, the strongest and weakest ones, and the ones around DC.

Table 4.2: Selected Subcarrier Frequencies and Channel Values ($\Delta f = 976.5625$ kHz)

k	f_k (MHz)	$\lvert H[k]\rvert$	$\angle H[k]$ (rad)
−128	−125.000000	0.3101	1.4725
−127	−124.023438	0.9240	1.1578
−45	−43.945312	0.0090	−0.4678
−14	−13.671875	1.8275	1.4827
−1	−0.976562	0.9592	0.0955
0	0.000000	0.7119	0.3598
1	0.976562	0.9891	0.4804
127	124.023438	0.4351	−0.4769

In the figure the stems sample the gray curve, which is the continuous $|H(f)|$ and $\angle H(f)$ of Q2(b) on the same axis. The bottom axis is the subcarrier index k, and the top axis of the magnitude panel gives the matching $f_k = k\Delta f$ in MHz. The subcarriers cover exactly one period $[-F_s/2, F_s/2)$ of $H(f)$, i.e. the middle half of the Q2(b) plot. Since $\Delta f = 977$ kHz is still about 5 times finer than the 5 MHz ripple, the stems follow every peak and null. The strongest subcarrier is $k = -14$ ($|H| = 1.83$) and the weakest is $k = -45$ ($|H| = 0.009$, almost a null). Neighboring subcarriers differ only a little, which is the per-subcarrier flatness mentioned in (b).

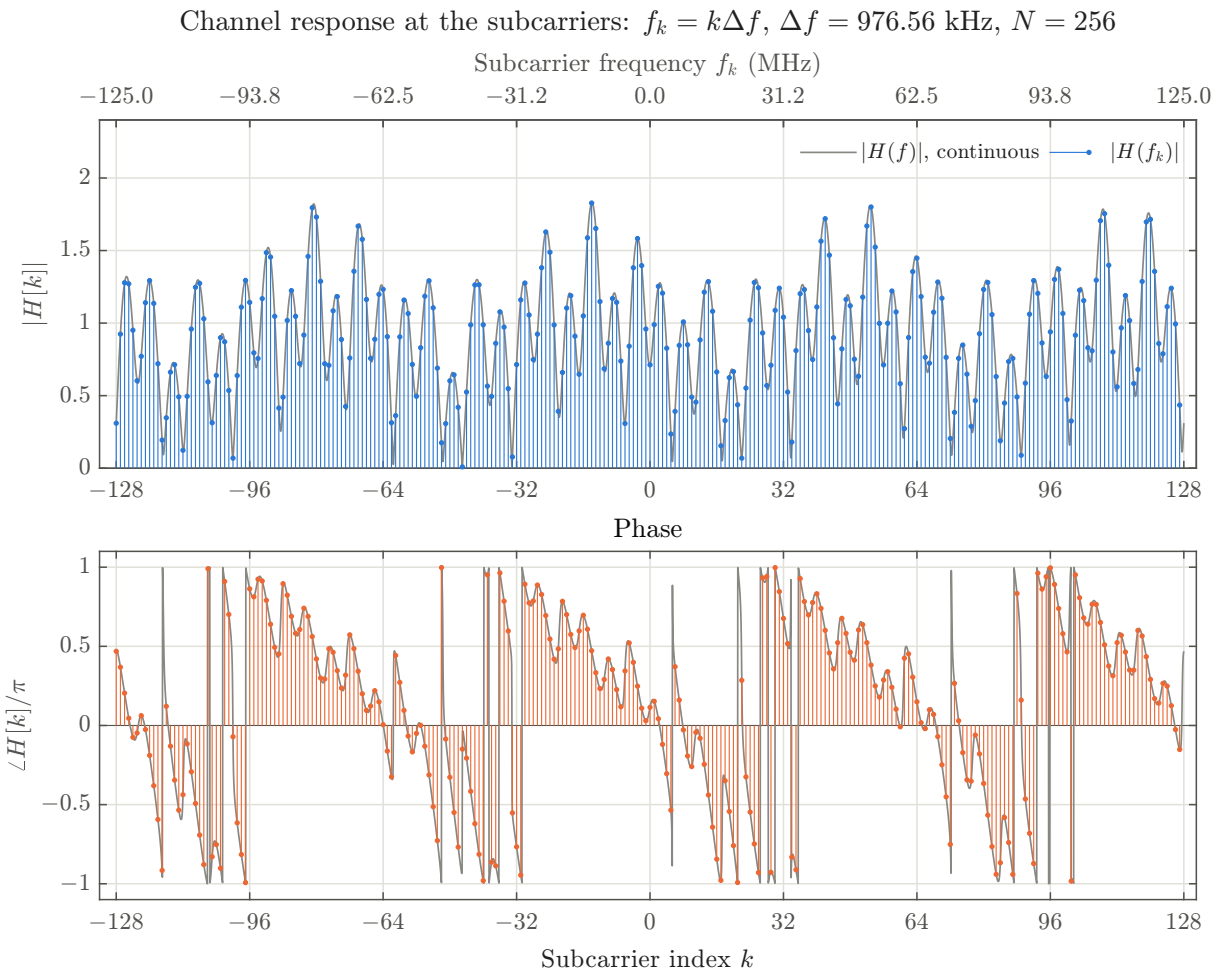


Figure 4.1: Magnitude and Phase of $H[k] = H(f_k)$ at the $N = 256$ Subcarriers (stems) with the Continuous $H(f)$ (gray)

Q4 — OFDM Transmission over the Channel

Now, let transmit the OFDM signal through the channel. Use the parameters calculated in Q3.

Everything in Q4 uses $N = 256$ and $N_g = 64$ from Q3, and the channel $h[n]$ of Q2 ($L = 51$).

5.1 (a) QPSK Data, IFFT and Cyclic Prefix

Please randomly generate frequency domain data from QPSK constellation (black constellation points in Homework 1). Draw the real part and imaginary part of frequency domain data $X[k]$ versus subcarrier index. Then transform it to time domain and add the cyclic prefix with length specified in Q3, which is indicated by $x[n]$. (10%)

The black points of HW1 Fig. 1 lie at 45°, 135°, 225° and 315° on the unit circle. As in HW1, the two bits (b_0, b_1) assigned to subcarrier k map to

$$X[k] = \frac{(1 - 2b_0) + j(2b_1 - 1)}{\sqrt{2}}, \tag{5.1}$$

i.e. $(0,1) \to e^{j\pi/4}$, $(1,1) \to e^{j3\pi/4}$, $(1,0) \to e^{j5\pi/4}$ and $(0,0) \to e^{j7\pi/4}$. I draw $2N = 512$ random bits with `randi`, continuing the seed-39 stream after the four γ_i, and put one symbol on each subcarrier $k = -128, \ldots, 127$. The four symbols occur 72, 66, 63 and 55 times. Every $X[k]$ has $|X[k]| = 1$, so its real and imaginary parts only take the values $\pm 1/\sqrt{2}$ (Figure 5.1). With 256 stems the full-band panels are dense, so the bottom row of that figure zooms into $k = -16 \sim 15$, where each symbol can be read off.

For the transform I use the unitary DFT pair

$$x[n] = \frac{1}{\sqrt{N}} \sum_{k=-N/2}^{N/2-1} X[k]\, e^{j2\pi kn/N}, \qquad Y[k] = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} y[n]\, e^{-j2\pi kn/N}, \tag{5.2}$$

implemented as `sqrt(N)*ifft(ifftshift(X))` and `fftshift(fft(y))/sqrt(N)`. This keeps the average power of $x[n]$ at 1, the same as $X[k]$; any matching pair of scalings would give the same $H[k]$ in (c). The CP is the last $N_g = 64$ samples copied to the front:

$$x_{\text{cp}}[n] = [\, x[N - N_g], \ldots, x[N-1],\ x[0], \ldots, x[N-1] \,], \qquad 320 \text{ samples}. \tag{5.3}$$

The question calls this CP-extended signal $x[n]$. I write $x_{\text{cp}}[n]$ for it so it is not confused with the IFFT output, and the figures and the convolution in (b) use $x_{\text{cp}}[n]$. In Figure 5.2 the gray block ($n = 0 \sim 63$) is the CP and the green block ($n = 256 \sim 319$) is where it was copied from; the difference between them is exactly 0. The time signal looks like noise with a peak of $|x| = 2.15$, as expected for 256 subcarriers with random phases.

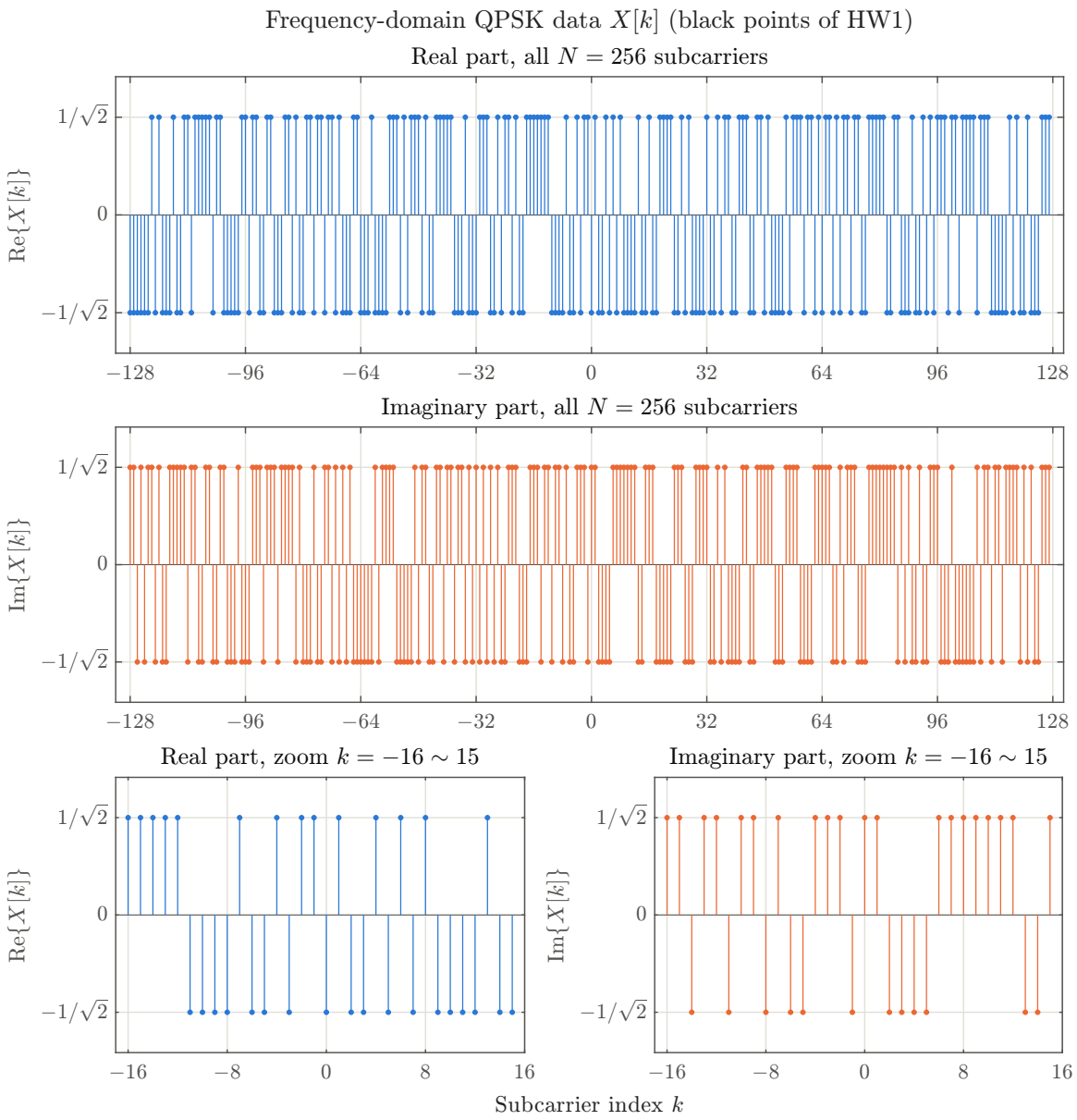


Figure 5.1: Real and Imaginary Parts of the QPSK Data $X[k]$ versus Subcarrier Index (bottom: zoom on $k = -16 \sim 15$)

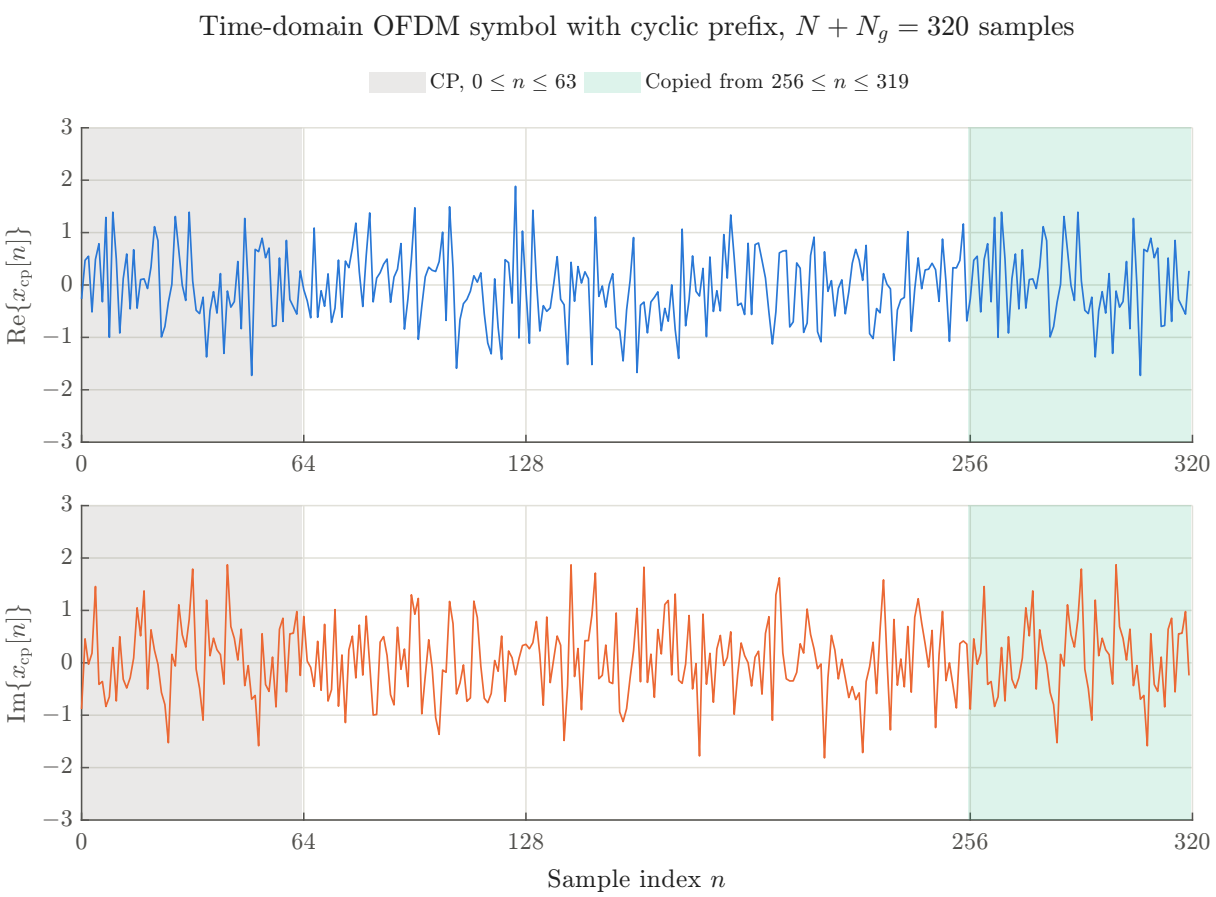


Figure 5.2: Real and Imaginary Parts of the Transmitted Signal with Cyclic Prefix, $x_{\mathrm{cp}}[n]$

5.2 (b) Received Signal

Use your own channel generated in Q2. Assume the discrete-time channel impulse response is described by

$$h[n] = \sum_{i=0}^{3} h_i \, \delta[n - n_i], \qquad n_i = \tau_i F_s.$$

Then generate the received signal $y[n] = h[n] \otimes x[n]$, where $\otimes$ denotes linear convolution. Then, draw the real-part and imaginary-part of your time-domain received signal first. Remove CP, transform it back to frequency domain by DFT to obtain $Y[k]$. Draw the magnitude of your frequency domain signal versus subcarrier index. (10%)

The transmitted signal here is $x_{\text{cp}}[n]$ from (a), so $y[n] = h[n] \otimes x_{\text{cp}}[n]$, and `conv(hn, xcp)` returns $51 + 320 - 1 = 370$ samples. In the figure below, the first 64 (gray) are the CP region and get removed. The last 50 (red) are the channel tail, i.e. echoes of the end of the symbol that arrive after the transmitter has stopped. In a continuous stream this tail would land on the CP of the *next* symbol, which is thrown away as well; this is how the CP blocks ISI. The receiver keeps $n = 64 \sim 319$ and takes the DFT.

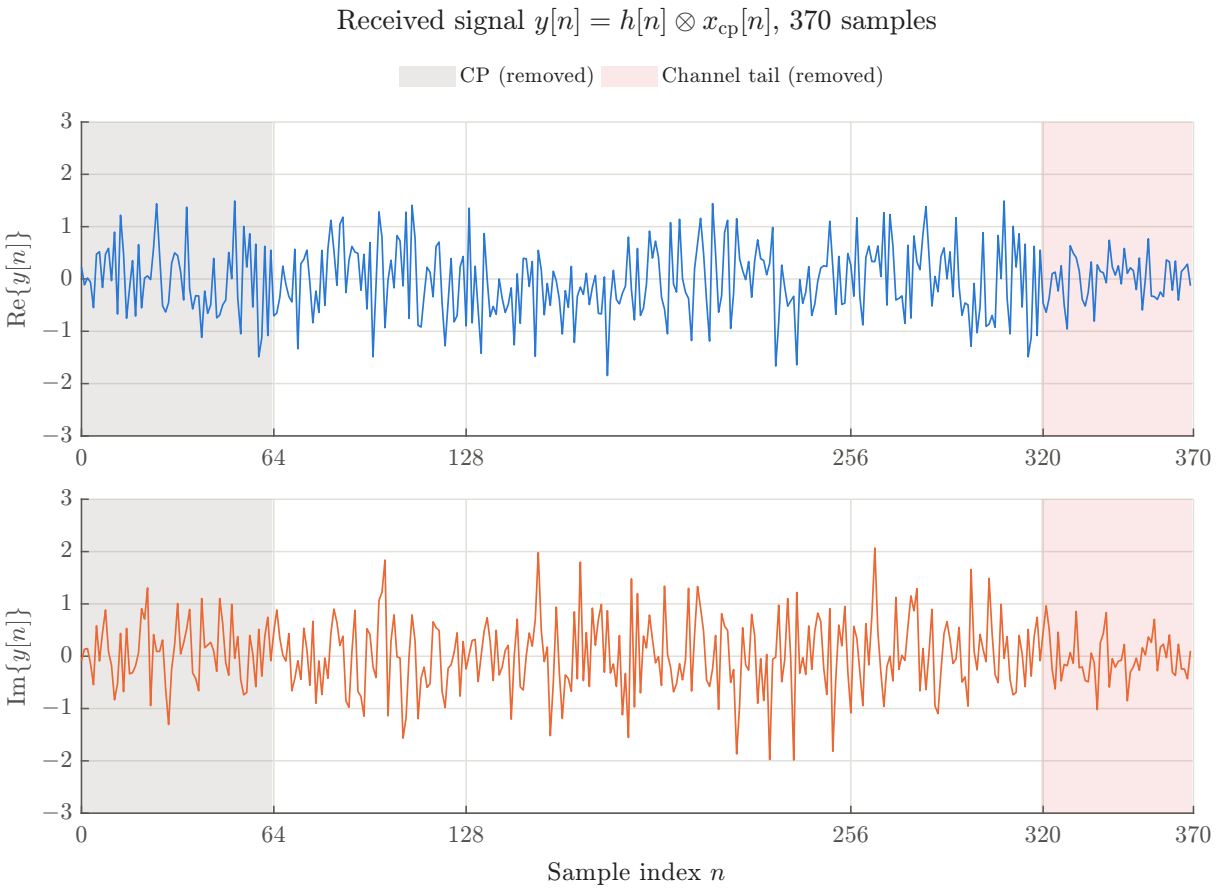


Figure 5.3: Real and Imaginary Parts of the Received Signal $y[n]$

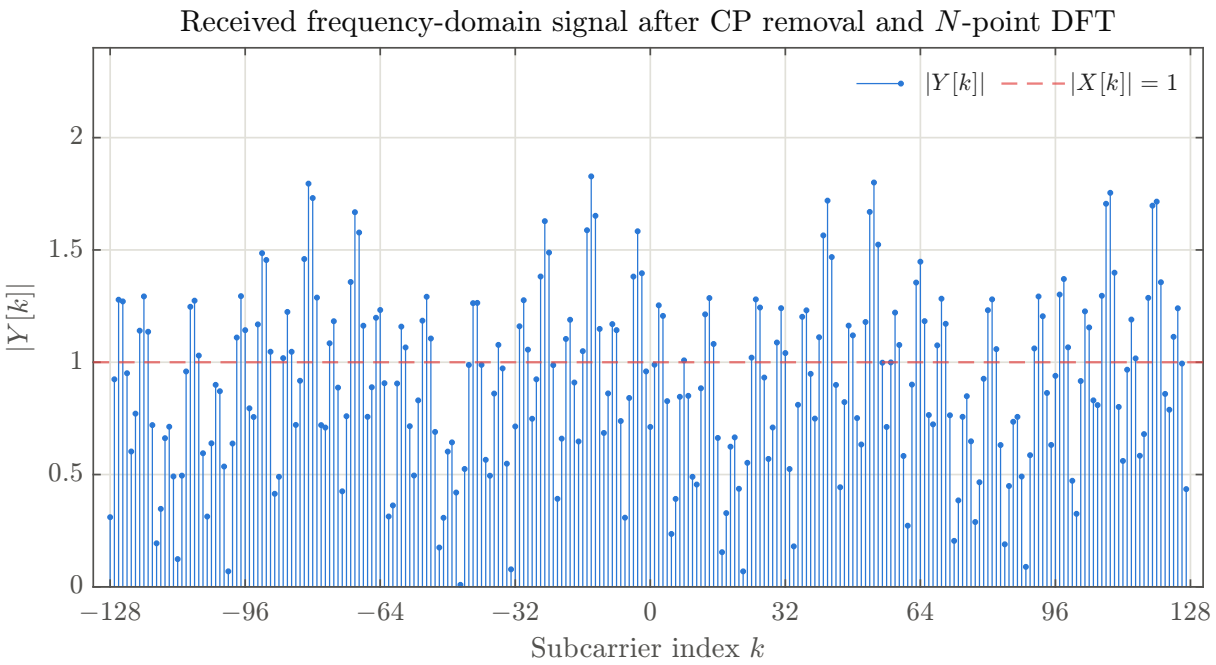


Figure 5.4: Magnitude of the Received Frequency-Domain Signal $Y[k]$

$|Y[k]|$ has exactly the shape of $|H[k]|$ in Q3(c). Part (c) shows that $Y[k] = H[k]X[k]$, and QPSK has $|X[k]| = 1$ (red dashed line), so $|Y[k]| = |H[k]|$; the data only rotates each subcarrier. Subcarriers in a deep fade, such as $k = -45$ with $|Y| = 0.009$, arrive almost erased. With noise they would fail first, which is why real OFDM systems spread channel coding across subcarriers. The mean of $|Y[k]|^2$ over the symbol is 1.000, which again matches $\sum_i |h_i|^2 = 1$.

5.3 (c) Channel Estimate $H[k] = Y[k]/X[k]$

Please calculate

$$H[k] = \frac{Y[k]}{X[k]}$$

Draw the magnitude of the frequency domain response $H[k]$ (10%). Observe this figure and the figure in Q3(c). Are they the same or not the same? Please explain the reason. (5%)

Dividing by the known data removes the QPSK phases. The top panel shows $|Y[k]/X[k]|$ as stems together with the Q3(c) values $|H(f_k)|$ as black circles (every 4th subcarrier, to keep the plot readable); the phase is shown below it.

They are the same. Over all 256 subcarriers the largest difference between $Y[k]/X[k]$ and $H(f_k)$ from Q3(c) is 7.9×10^{-15}, which is only rounding error. Two things have to hold for this.

First, the CP turns the linear convolution into a circular one. The kept samples are

$$y[N_g + n] = \sum_{l=0}^{L-1} h[l]\, x_{\text{cp}}[N_g + n - l], \qquad n = 0, \ldots, N-1. \tag{5.4}$$

Because $l \le L - 1 = 50 \le N_g = 64$, the index $N_g + n - l$ never drops below 0. When it falls inside the CP, the CP supplies $x[(n - l) \bmod N]$, which is the value a circular convolution would use. Hence

$$y[N_g + n] = \sum_{l} h[l]\, x[(n - l) \bmod N] = (h \circledast x)[n], \tag{5.5}$$

and the script confirms it (difference 8.9×10^{-16}). By the DFT convolution theorem, $Y[k] = H_N[k]\, X[k]$ with $H_N[k] = \sum_n h[n] e^{-j2\pi kn/N}$, the N-point DFT of $h[n]$. Each subcarrier is multiplied by one complex number and nothing leaks in from the other subcarriers.

Second, the DFT of $h[n]$ has to equal $H(f)$ at the subcarriers. It does, because the delays lie exactly on the sample grid ($\tau_i = n_i/F_s$ with integer n_i):

$$H_N[k] = \sum_i h_i\, e^{-j2\pi k n_i/N} = \sum_i h_i\, e^{-j2\pi (kF_s/N)\tau_i} = H(f_k). \tag{5.6}$$

Together, $Y[k]/X[k] = H(f_k)$ exactly. The simulation has no noise and no timing or frequency offset, and the channel does not change during the symbol, so nothing else can pull the two apart.

To see when they would differ, the bottom panel reruns the whole chain with shorter CPs. With $N_g = 50$ and 64 the error stays at the rounding level. With $N_g = 49$ the 200 ns path reaches one sample beyond the CP and the error jumps to 1.54×10^{-2} on every subcarrier; only one received sample is wrong, so the error is an impulse in time and flat across k. With $N_g = 32$, 18 samples of the last path fall outside the CP and the error grows to 0.25. The circular-convolution property is gone, energy leaks between subcarriers (ICI), and $Y[k]/X[k]$ no longer equals the channel response. This confirms the bound $N_g \ge L - 1 = 50$ from Q1(d) and Q3(a). The two figures would also differ if the path delays were not multiples of T_s, because then the sampled channel spreads over many taps and $H_N[k]$ only approximates $H(f_k)$. With receiver noise, Y/X would become a noisy estimate, worst at faded subcarriers such as $k = -45$.

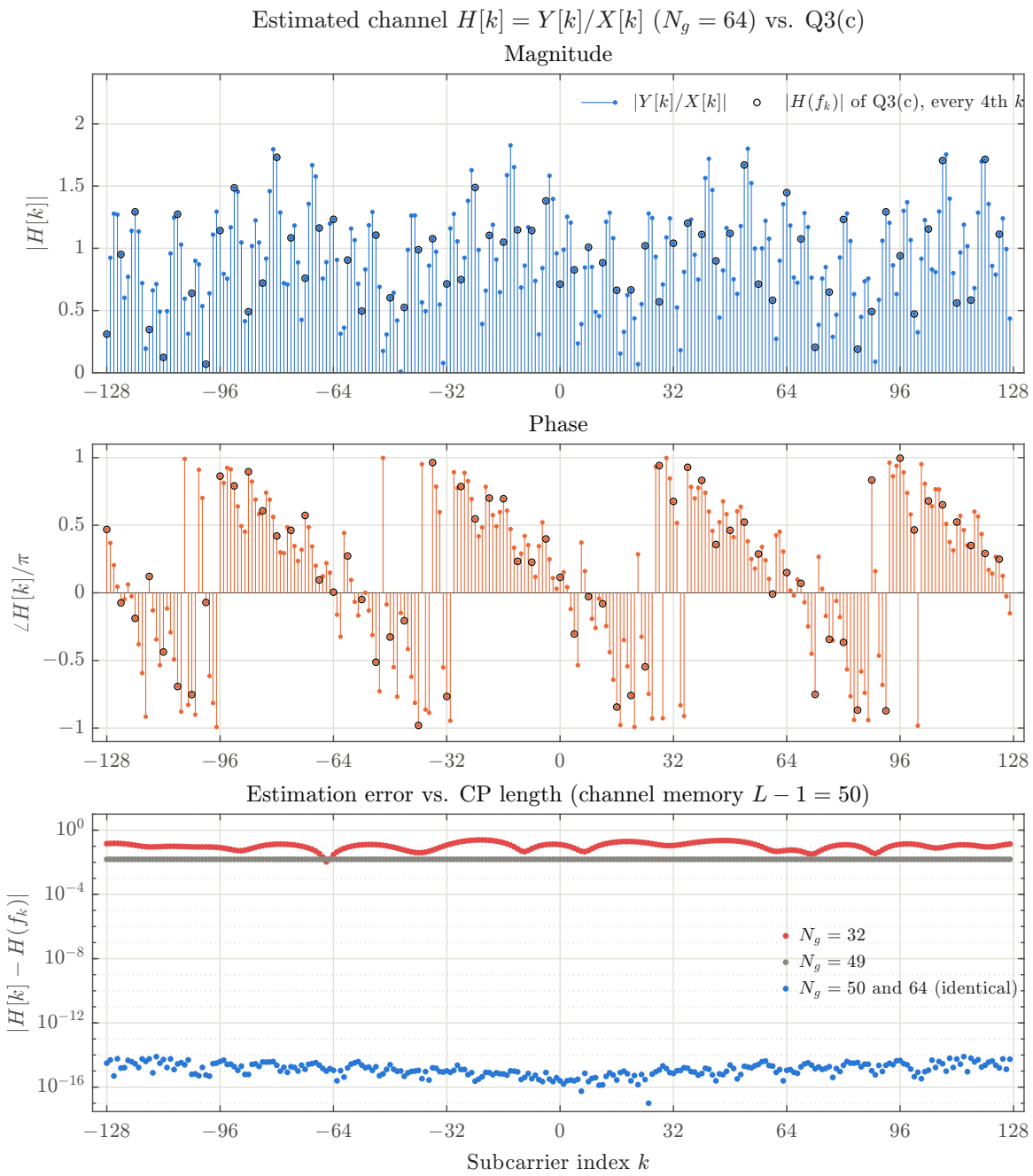


Figure 5.5: Estimated $H[k] = Y[k]/X[k]$ (stems) vs. $H(f_k)$ from Q3(c) (circles), and the Estimation Error for Different CP Lengths

AI Tool Disclosure

Please disclose how you use AI tools and its version.

I used **Claude Opus 5.5** to help with this homework in the following four parts:

(1) Understanding the homework specification.

(2) Assisting with the MATLAB code.

(3) Plotting the figures.

(4) Writing this LaTeX report.